Predicting Bicycle Theft Risk Using Big Data

Lambda Architecture + Real-Time Machine Learning

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MBAI 5100G - Big Data Systems Design

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Key Insights

Nighttime Risk

Evening and night hours show the highest theft activity with 8,613 incidents during nighttime

Apartment Targeting

Multi-unit residential buildings like apartments and condos are the most frequently targeted locations

Downtown Hotspot

Waterfront communities and downtown areas represent major concentration zones for bicycle theft

Private Property Risk

Private spaces including apartment buildings and private property structures show higher risk than public areas

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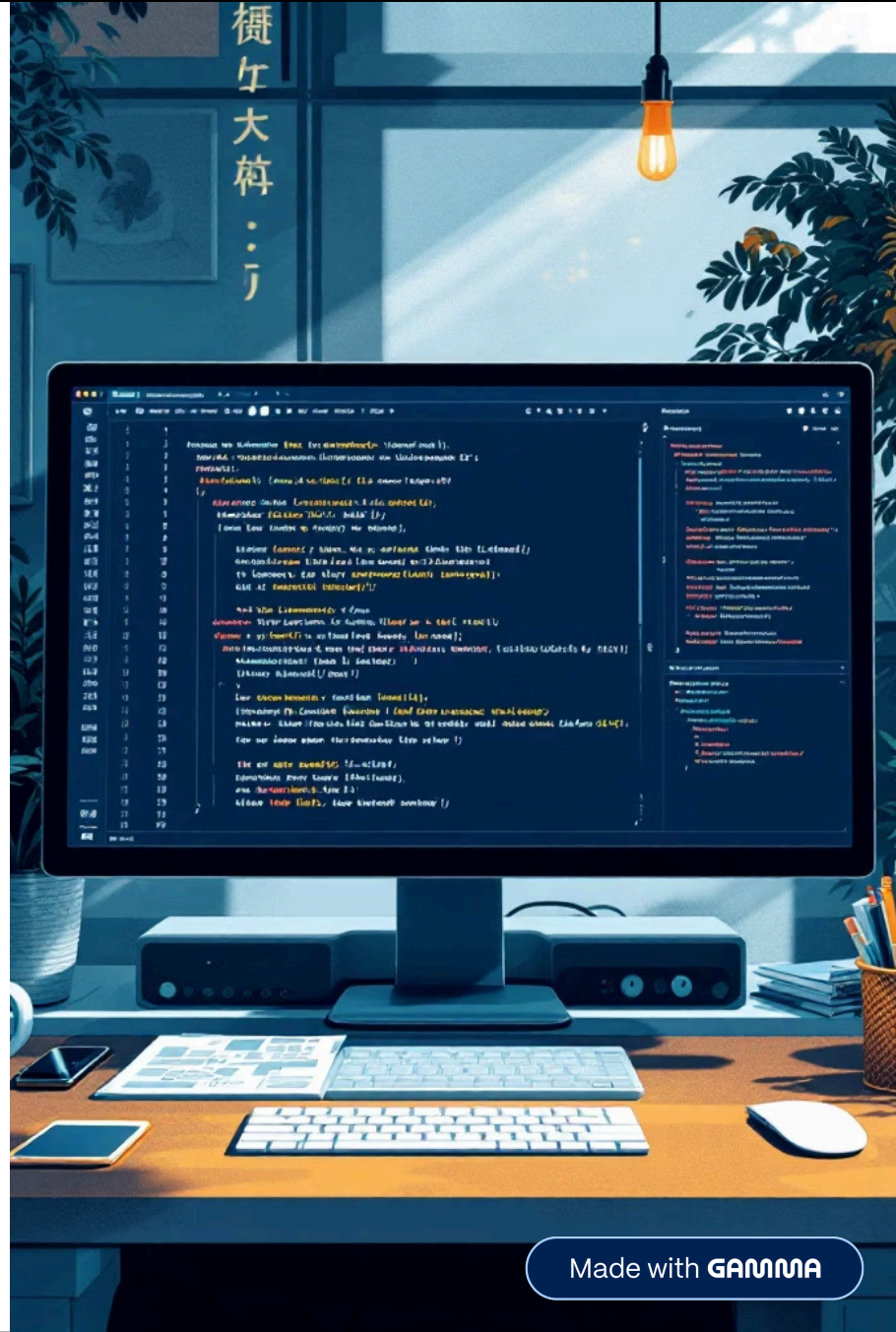
The Data

1	**Temporal** Patterns across the day and night reveal when activity is most concentrated.
2	**Spatial** Property and place types show the settings most often associated with incidents.
3	**Geospatial** Specific areas help identify where incidents cluster within the city.



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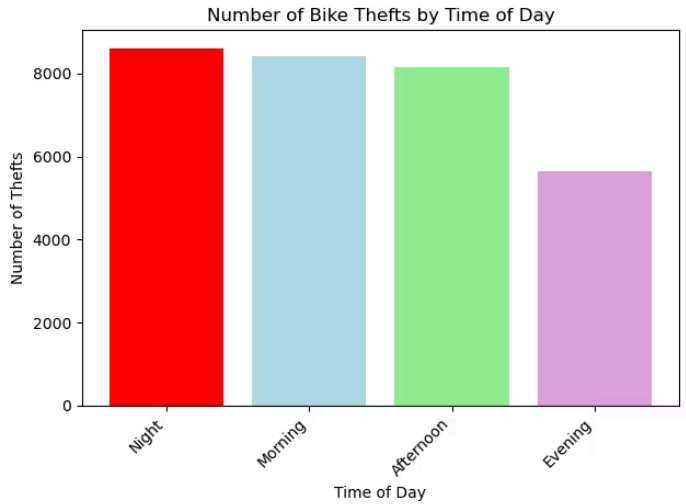

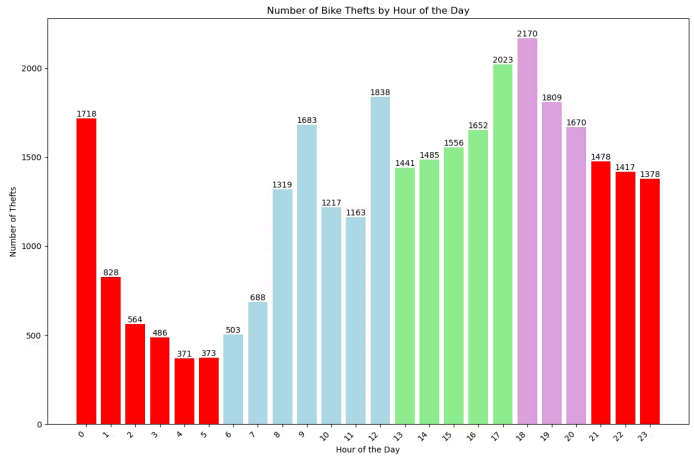

Time Distribution

1	**Night** **8,613** incidents
2	**Morning** **8,411** incidents
3	**Afternoon** **8,157** incidents
4	**Evening** **5,649** incidents

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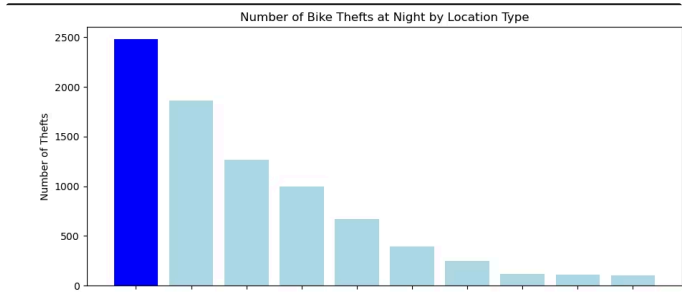

Location Breakdown

1	**Apartment** **2,479** incidents
2	**Single Home** **1,865** incidents Rooming House, Condo
3	**Streets & Roads** **1,265** incidents Highways, Bicycle Path, Private Road
4	**Private Property Structure** **995** incidents Pool, Shed, Detached Garage
5	**Parking Lots** **673** incidents Apt., Commercial Or Non-Commercial

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Top Theft Hotspots

1	St Lawrence-East Bayfront-The Islands **166 incidents**
2	Wellington Place **164 incidents**
3	Harbourfront-CityPlace **135 incidents**
4	Annex **130 incidents**
5	Fort York-Liberty Village **121 incidents**

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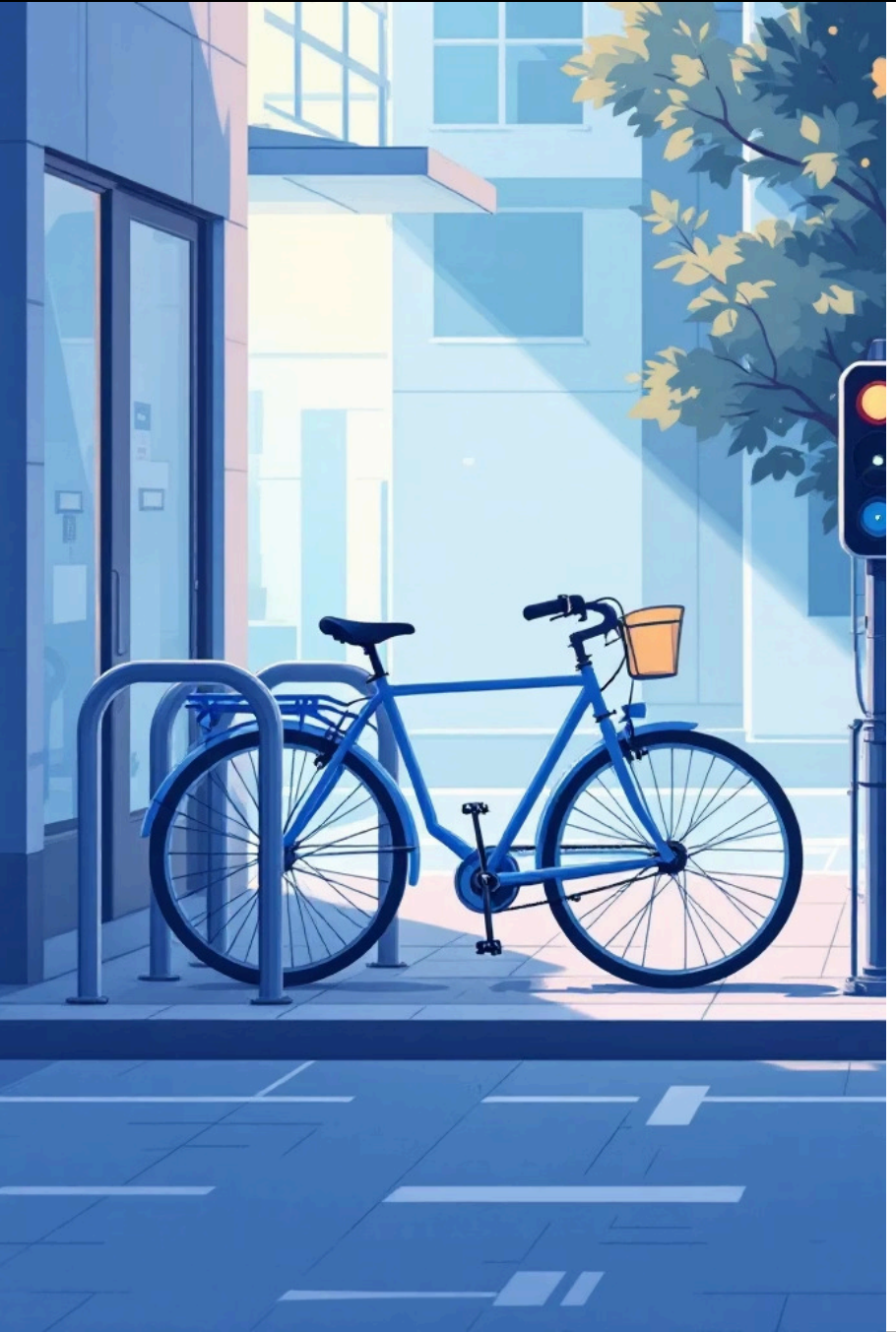

Mitigating Bicycle Thefts

City Response

Increase police patrols at apartment complexes between 6 PM and midnight, focusing on identified hotspots

Resident Action

Secure bicycles properly using quality locks and store in secure locations overnight

Technology Solution

Use "Bike Guard" app to predict real-time risk levels based on bike details

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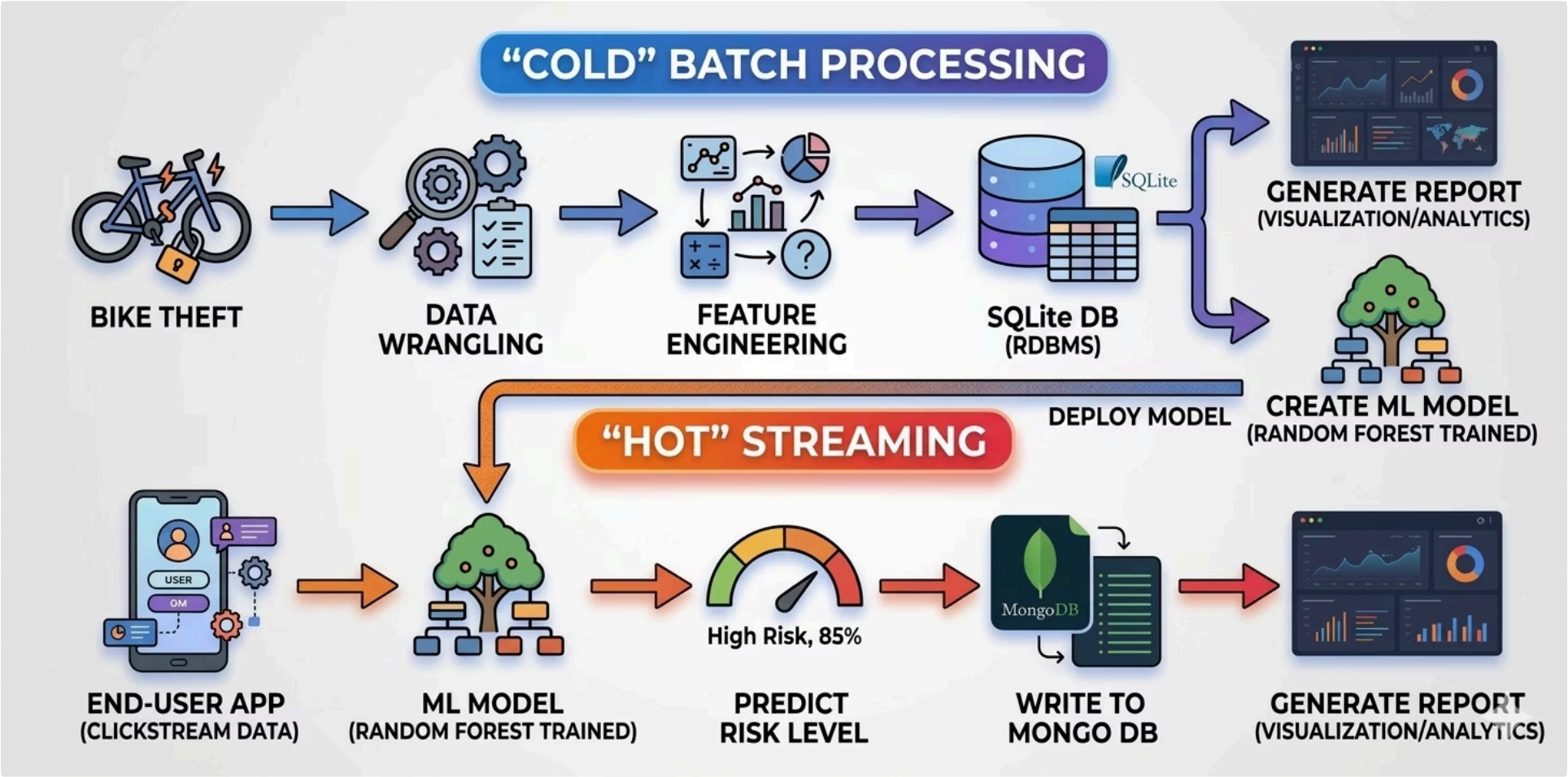
Big Data Pipeline



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Conclusion

Predictive Power Unleashed

Big data and machine learning are indispensable tools for uncovering and forecasting intricate bicycle theft patterns.

Integrated Processing for Impact

Combining batch and real-time data processing delivers both comprehensive historical insights and immediate, actionable intelligence.

Enhanced Bike Safety

These advanced analytical capabilities translate directly into practical applications for improving city-wide bicycle security and resident safety.

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Questions & Answers